

Sushmita Paul

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RESEARCH INTERESTS

SLM Reasoning & Evaluation, Reliable AI, Efficient ML Systems, Computer Vision

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering (Honors)

Feb 2020 – Mar 2025

Dhaka, Bangladesh

- **CGPA:** 3.80/4.00 | **Award:** Dean's List, BUET
- **Selected Coursework:** Artificial Intelligence, Machine Learning, Operating Systems, Data Structures & Algorithms.

RESEARCH EXPERIENCE

Diagnosing Small Language Models under Incremental Observations

Research Collaboration with Prof. Ch. Md. Rakin Haider [Project]

May 2025 – Present

BUET

- Study how **small language models keep track of task state** during long, multi-turn interactions, and whether verified external memory makes their reasoning more reliable.
- Use interactive breadth-first and depth-first graph traversal as controlled test cases, comparing raw prompts, prose and structured reminders, **state-report verification and correction**, and retained versus cleared history across Qwen and Gemma.
- Built a reproducible **vLLM/Slurm evaluation pipeline** with fixed data splits, resumable runs, and trajectory logs to measure completion, traversal correctness, invalid actions, state-report errors, and recovery after feedback.

Detection of Dengue Breeding Sites in Unplanned Urban Landscapes with Deep Learning from Remote Sensing

Undergraduate Thesis with Prof. Anindya Iqbal [Thesis]

Nov 2023 – Jan 2025

BUET

- Built a **two-stage framework** that detects breeding objects, segments buildings, and combines both signals through spatial containment; it reached **83.6% balanced accuracy**, 14.1 points above a joint detection-classification baseline.
- Analyzed how multiple rooftop objects change the chance of identifying a risky building, finding a probability above 0.8 when at least four objects are present even though individual class mAP50 was below 0.36.
- Curated a UAV imagery dataset with **5,845 objects across six classes and 289 building boundaries**, then compared YOLOv9 GELAN-C, SegGPT, YOLOv8m, and YOLOv11m.
- Wrote sliding-window postprocessing with modified Non-Maximum Merging to rebuild buildings across overlapping image tiles, and validated 50% overlap through sensitivity analysis.
- Turned the model outputs into a spatial inspection strategy with 89.1% true-positive and 78.1% true-negative rates, reducing estimated **inspection labor costs by 35%**.

MANUSCRIPTS & PRESENTATIONS

1. **Paul, S.**, et al. “Detection of Dengue Breeding Sites in Unplanned Urban Landscapes with Deep Learning from Remote Sensing.” Manuscript under review at *BMC Public Health* (Springer Nature). *Co-first author*.
2. **Paul, S.**, et al. “Identification of Dengue Breeding Sites from Point-Cloud Images.” *BEAR Summit 2025*, Bangladesh. Digital poster presentation.

SELECTED RESEARCH PROJECT

Medical Image Analysis: Lung Nodule Detection & Segmentation

Jan 2025

Machine Learning Course Project [[Code](#)]

BUET

- Trained SAM2 and SwinUNETR to outline lung nodules and Faster R-CNN to detect them in LIDC-IDRI CT scans, using hybrid Dice/BCE loss, mixed precision, and gradient accumulation.
- Reached **Dice 0.93, precision 0.96, and recall 0.89**, and reconciled the dataset’s multiple annotations into consensus ground truth.

PROFESSIONAL EXPERIENCE

Therap BD Ltd

Apr 2025 – Present

Machine Learning Engineer

Dhaka, Bangladesh

- Built LLM pipelines that turn clinical notes into concise summaries, combining healthcare data extraction, prompt design, hallucination checks, multi-model evaluation, and adversarial testing.
- Compared vision-language and segmentation models for deployment, built a real-time face detector, and sped up NVIDIA Jetson inference with TensorRT quantization.
- Compared encoder and decoder models for text classification, balancing accuracy against deployment cost and latency.

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, SQL

ML Frameworks & Methods: PyTorch, scikit-learn, Hugging Face, ONNX, Optuna, GANs, VAEs, diffusion models

Language Models: LoRA/QLoRA fine-tuning, vLLM, reasoning evaluation, prompt engineering, vision-language models, ModernBERT

Computer Vision: SAM/SAM2, YOLO, SwinUNETR, Faster R-CNN, OCR

Systems: Slurm, Linux, Git, Docker, Azure, NVIDIA Jetson, TensorRT, FP16/INT8 quantization, Celery, RabbitMQ

TEACHING & OUTREACH

- Tutored Higher Secondary Certificate students in Mathematics and Physics; delivered a technical seminar at Islamic University of Technology (IUT).